NON-NETWORK INTENSIVE RECORD KEEPING SYSTEM FOR SELF HELP GROUPS

20CJPJ1001 - PROJECT PHASE – II

Submitted by

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BONAFIDE CERTIFICATE

Certified that this Report titled **"Non-Network Intensive Record Keeping System for Self Help Groups"** is the bonafide work of **VAISHNAVI S (412521553060) , PAVITHRA R (412521553032)** who carried out the **20CJPJ1001 PROJECT PHASE - II** work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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A successful man is one who can lay a firm foundation with the bricks others have thrown at him.
— *David Brinkley*

Such a successful personality is our beloved founder Chairman, **Thiru. MJF. Ln. LEO MUTHU.** At first, we express our sincere gratitude to our beloved Chairman through prayers, who in the form of a guiding star has spread his wings of external support with immortal blessings.

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ABSTRACT

Self-Help Groups (SHGs) have emerged as pivotal instruments for promoting financial inclusion, women's empowerment, and sustainable rural development in India. These grassroots groups, typically consisting of 10 to 20 members, pool resources and provide microcredit and savings facilities to their members. Despite their widespread presence and growing significance, SHGs face considerable operational challenges due to reliance on Self-Help Groups (SHGs) play a crucial role in promoting financial inclusion and women's empowerment in rural India by enabling collective savings and microcredit activities. However, many SHGs still depend on traditional paper-based record-keeping systems which are prone to errors, data loss, and inefficiencies, particularly in regions with unreliable internet connectivity.

To overcome these challenges, the SHG Connect project introduces an offline-first mobile application tailored specifically to the needs of SHG members and facilitators. The application enables secure, uninterrupted local recording of transactions such as savings, loans, and repayments on the device, with automatic synchronization to a centralized cloud server when connectivity becomes available. Employing conflict-aware synchronization protocols, embedded databases, and a simple multilingual interface designed for semi-literate users.

SHG Connect enhances transparency, accuracy, and operational efficiency while safeguarding sensitive data via role-based access controls. By bridging the digital divide with a robust, scalable, and user-centric solution, this project supports transparent financial management, timely compliance reporting, and strengthens institutional maturity within the SHG ecosystem, aligning with government initiatives like DAY-NRLM to foster inclusive rural development and sustainable community empowerment.

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CHAPTER	ABBREVIATIONS	FULL FORM
1	SHG	Self-Help Group
2	DAY-NRLM	Deendayal Antyodaya Yojana – National Rural Livelihoods Mission

1. INTRODUCTION

1.1 BACKGROUND OF THE STUDY :

Self-Help Groups (SHGs) have emerged as a vital socio-economic platform for empowering marginalized rural populations, especially women, in India. Typically consisting of 10 to 20 members, SHGs focus on collective savings, credit access, and community-based development activities. Through pooling of resources and mutual support, members gain enhanced financial inclusion and an opportunity to improve their livelihoods. Government initiatives such as the Deendayal Antyodaya Yojana - National Rural Livelihoods Mission (DAY-NRLM) have further propelled the growth and institutionalization of SHGs across rural India. Despite their rapid proliferation and demonstrated impact, a significant operational challenge persists—the lack of efficient, reliable, and accessible digital tools for accurate record keeping, especially in regions with unreliable network connectivity. Many SHGs continue to depend on manual, paper-based registers, which are prone to errors, loss, and delay in reporting, thereby limiting transparency and decision-making effectiveness.

1.2 PROBLEM STATEMENT :

Self-Help Groups (SHGs) in India play a vital role in fostering financial inclusion, women's empowerment, and rural development. However, many SHGs grapple with significant operational challenges that hinder their effectiveness and sustainability. One of the primary issues is the lack of financial literacy among members, which results in poor management of savings, loans, and expenditure records, leading to frequent defaults and misappropriation. Additionally, SHGs face difficulties in accessing formal credit due to inadequate documentation, poor credit histories, and limited relationships with financial institutions.

Furthermore, operational inefficiencies such as weak governance, poor record-keeping, and dependence on manual processes increase the risk of data inaccuracies and mismanagement. In rural areas with unreliable internet connectivity, conventional digital management tools are ineffective, forcing SHGs to rely on traditional paper documentation, which is prone to errors and loss. These problems collectively undermine transparency and accountability, obstruct timely reporting, and inhibit growth. Addressing these issues through a technology-driven, offline-capable management system is crucial to strengthening the financial health, operational efficiency, and long-term sustainability of SHGs across India.

1.3 OBJECTIVE OF THE PROJECT:

The main objective of the SHG Connect project is to develop an offline-first mobile application specifically designed to meet the unique needs of rural Self-Help Groups, enabling them to manage financial transactions efficiently despite limited internet connectivity. The specific objectives include:

- To create a secure and reliable platform for SHG members to record savings, loans, repayments, and group activities locally on their mobile devices without requiring continuous internet access.
- To implement intelligent background synchronization that automatically uploads stored data to a centralized server when network connectivity is available, resolving data conflicts transparently.
- To design an intuitive, multilingual, and icon-driven user interface tailored for semi-literate users to facilitate broad adoption within rural communities.
- To ensure data security and privacy through role-based access control, encrypted data storage, and secure transmission protocols.
- To support federation officers and administrators with centralized data aggregation, monitoring dashboards, and reporting tools to facilitate effective governance and decision-making.
- To build a scalable and extensible system architecture capable of serving large numbers of SHGs across diverse geographic and linguistic regions.

1.4 SCOPE OF THE PROJECT:

The scope of the SHG Connect project encompasses the design and development of an offline-first mobile application tailored specifically for Self-Help Groups in rural India. The core focus is to enable SHG members to effectively record financial transactions such as savings, loans, repayments, and group activities locally without dependence on continuous internet connectivity. Embedded databases like SQLite ensure uninterrupted data persistence, while intelligent synchronization modules upload data securely when connectivity becomes available, resolving conflicts to preserve data accuracy. The application will feature a user-friendly, multilingual, and icon-driven interface optimized for semi-literate users, promoting wide adoption across diverse linguistic regions.

In Scope

- Offline transaction recording with local data storage.
- Automatic, background data synchronization with centralized cloud storage.
- Conflict-aware synchronization protocols ensuring data integrity.
- Multilingual, icon-based user interface for low literacy users.
- Automated generation of financial and compliance reports in PDF and Excel formats.
- Federation-level data aggregation and dashboard reporting for administrators.
- Scalable architecture to support many SHGs across varied geographies.

Out of Scope

- Web and desktop portals for SHG management.
- Voice-based data entry and biometric authentication.
- Direct integration with government databases or portals.
- Advanced analytics, machine learning features.
- Extended training modules and offline voice assistance.

Project Impact

The SHG Connect solution aims to revolutionize grassroots financial management by empowering rural women with accessible, dependable tools for transparent record-keeping and easy data synchronization. This will enhance the operational efficiency, credibility, and decision-making of SHGs while fostering greater trust among members. By facilitating compliant reporting and improved data aggregation, the project will support stronger oversight and governance at federation levels. Ultimately, the system will contribute to advancing the DAY-NRLM mission of inclusive growth and women's empowerment through innovative technology solutions that bridge the digital divide in rural India.

2. LITERATURE REVIEW

2.1 INTRODUCTION:

Digital transformation has been a catalyst in revolutionizing financial inclusion efforts globally, and Self-Help Groups (SHGs) stand at the forefront of this movement in rural India. SHGs primarily serve as grassroots institutions that empower marginalized women through collective savings, credit access, and social development. However, despite government initiatives and the rise of mobile financial applications, many SHGs continue facing challenges in maintaining accurate, timely, and accessible financial records due to infrastructural limitations such as intermittent internet connectivity and low digital literacy among members.

Several studies have explored offline-first mobile applications, decentralized data synchronization, and user-friendly interfaces designed for low-literacy users to overcome these challenges. Research highlights the importance of embedding conflict-aware synchronization protocols that ensure consistency and reliability of shared financial data in collaboration-driven environments.

This literature review first summarizes key research in this domain, comparing methodologies, advantages, and constraints. Next, it discusses the advantages and limitations of existing systems, focusing on their applicability to rural SHGs. Finally, the review identifies gaps in current offerings, motivating the need for innovative offline-first, group-oriented, and user-centric solutions such as the SHG Connect project.

2.2 LITERATURE REVIEW

1. Offline Support of Android Applications for Mobile Sensing

Shigemoto et al. proposed an Offline Support of Android Applications for Mobile Sensing **[1]** that combines local caching with automatic background synchronization to ensure continuity of data collection in low-connectivity environments. Their system first stores sensor readings on the device and later pushes them to a remote server once connectivity is restored, thereby preventing data loss during outages or application crashes. The authors demonstrate through field experiments that this offline-first architecture maintains data integrity over long periods, even when devices experience intermittent or poor network coverage.

The study is particularly relevant for rural and mission−critical deployments because it validates the practicality of delayed synchronization and robust on-device persistence. At the same time, Shigemoto et al. acknowledge trade−offs such as increased battery consumption caused by frequent background sync operations. The work also focuses primarily on unidirectional sensor data uploads and does not deeply explore multi-user scenarios, complex transaction models, or conflict resolution when several clients update overlapping records, which are essential concerns in SHG financial applications.

2. Helping Hand: Interactive Mobile Application for Underprivileged Assistance

Chowdhury et al. presented Helping Hand: Interactive Mobile Application for Underprivileged Assistance **[2]** designed to streamline assistance for underprivileged users by integrating donation, beneficiary selection, emergency help, and rating features. The authors emphasized that the system supports both offline data capture and later synchronization, enabling users to register requests and donations even when network access is unavailable, which are then reconciled with the server when connectivity returns. Their results indicated that bringing multiple social-service workflows into a single mobile platform can reduce process friction, increase transparency, and build trust between donors and beneficiaries.

However, the work primarily addressed social aid and emergency support, not structured group-based financial management; consequently, it did not introduce detailed models for SHG-style ledgers, role-based access among members and leaders, or robust conflict resolution for overlapping financial records. The system does not model SHG-style group ledgers, role-based access hierarchies, or detailed reconciliation of concurrent financial updates submitted by multiple actors. Consequently, while the work provides valuable lessons on usability and offline capture, it offers limited guidance on designing conflict-aware, multi-actor transaction systems like those needed for SHG Connect.

3. Mobile-Based Location Tracking without Internet Connectivity

Bani Hani et al. proposed a Mobile-Based Location Tracking without Internet Connectivity **[3]** solution that operates without continuous internet by employing a custom latitude longitude algorithm and storing data locally on the device. The authors demonstrated that essential safety-related information such as user coordinates can be collected and preserved entirely offline, with optional uploading to the cloud when connectivity becomes available. Nevertheless, the study was limited to geospatial tracking and did not consider transactional data models, multi-user group interactions, or complex synchronization semantics, meaning that its techniques cannot directly handle rich SHG financial workflows or concurrent edits by multiple stakeholders.

The study illustrates how core system functionality can be designed to be independent of real-time connectivity and still remain effective and resilient. However, the scope of the work is limited to geospatial tracking, with no consideration of complex transactional data models, shared ledgers, or collaborative workflows involving multiple users. It does not discuss mechanisms for resolving conflicts between offline updates from different devices, nor does it address domain-specific requirements such as financial integrity, role separation, or audit trails that are central to SHG-oriented record-keeping platforms.

4. Progressive Web Apps to Support Critical Systems in Low or No Connectivity Areas

Josephe et al. discussed the use of Progressive Web Apps to Support Critical Systems in Low or No Connectivity Areas **[4]** enhanced with service workers and SMS-based fallback channels to keep critical systems operational under low or no connectivity conditions. Their architecture allows web applications to cache essential resources locally and, when networks fail, to fall back to SMS as a lightweight transport layer for transmitting key updates, thus sustaining minimum service continuity. The authors argued that this strategy can significantly reduce dependence on mobile data and provide resilience for critical applications in constrained environments. However, they also noted practical limitations, including the small payload size of SMS messages and reliance on browser capabilities, which restrict the complexity and volume of data that can be exchanged. In addition, the solution was not tailored to group-centric financial record-keeping; it did not address structured ledger formats, hierarchical roles, or high-volume transaction histories typical of SHGs.

This work highlights an alternative pathway for designing resilient applications that must survive harsh connectivity constraints, emphasizing compatibility across devices through web technologies. However, practical limitations such as restricted payload sizes in SMS and reliance on browser capabilities place constraints on the volume and complexity of data that can be handled. The solution is not tailored for structured group financial workflows: it does not address persistent ledger structures, rich transaction histories, or the role-based access and multi-device conflict handling that SHG digitisation requires.

5. Offline Data Sharing on Android Application

Zhang et al. proposed an Offline Data Sharing on Android Application **[5]** that leverages local Wi-Fi hotspots, Bluetooth, and a lightweight NanoHTTPD server to enable peer-to-peer transfer of large files between devices. The authors showed that documents, images, and other sizeable artifacts can be exchanged without internet connectivity or centralized infrastructure, which is advantageous in resource-constrained rural contexts. Their experiments indicated that this approach maintains data availability and integrity during transfers and can be used to distribute content such as forms or reports locally.

Despite these strengths, the method depends on hardware features like Bluetooth and hotspot support on participating devices and becomes less efficient as the number of recipients grows. Moreover, the work focused on file distribution rather than transactional synchronization; it did not incorporate end-to-end security controls, user roles, or mechanisms for merging concurrent changes elements that are essential in a production-grade SHG record-keeping platform.

2.3 COMPARATIVE REVIEW OF EXISTING RESEARCH

S.NO	CRITERIA	Existing System	Proposed System (SHG Connect)
1.	Offline-first support	1. Focuses on mobile sensing data with local caching and automatic background sync in low-connectivity environments. 2. Ensures data persistence across crashes and long offline periods but mainly handles one-way uploads. 3. Does not address structured financial records, multiple user roles, or conflict resolution between concurrent updates.	1. Adopts the offline-first principle for SHG savings, loans, and meetings with reliable local storage and delayed sync. 2. Supports two-way data flow where devices both send and receive updated group records from a central server. 3. Adds explicit conflict-aware synchronization and role-based permissions tailored to SHG workflows.
2.	Synchronization and conflicts	1. Integrates donation, beneficiary selection, and emergency help into a single app for underprivileged support. 2. Allows offline capture of requests and donations with later synchronization to reduce delays. 3. Centred on one-to-one aid workflows, without group ledgers, periodic financial summaries,or SHG governance structures.	1. Integrates multiple SHG functions (savings, loans, attendance, meetings, expenses) into one unified platform. 2. Uses offline capture plus sync specifically for recurring SHG transactions, not just occasional donations. 3. Provides structured group ledgers, monthly/meeting reports, and leader oversight for accountability.

3.	Data model and domain fit	**1.** Tracks user location using a custom algorithm and stores coordinates offline on the device. **2.** Demonstrates that critical safety data can be collected without continuous internet. **3.** Limited to single-user geolocation data, with no multi-user collaboration, financial models, or detailed audit trails.	**1.** Extends offline design principles from simple coordinates to complex, multi-field SHG transactions and member records. **2.** Ensures continuity of financial operations and meeting logging even in complete network outages. **3.** Maintains audit-ready histories for each member and group, supporting reviews, corrections, and compliance.
4.	User roles and governance security and trust	**1.** Uses Progressive Web Apps with service workers and SMS fallback to keep critical services running during outages. **2.** Caches core resources in the browser and uses SMS for minimal data exchange when networks fail. **3.** Constrained by SMS payload size and browser dependence; not optimized for heavy, structured financial data.	**1.** Uses mobile-app approach with a full offline database, avoiding SMS limits and browser dependence. **2.** Supports rich financial ledgers, attachments (photos, passbooks), and detailed metadata fully offline. **3.** Still embodies resilience to poor networks but in a way aligned with SHG requirements (reports, passbooks, role-based access).
5.	Reporting and decision support	**1.** Enables peer-to-peer sharing of large files via Wi-Fi hotspot,Bluetooth, and a lightweight local web server.	**1.** Uses centralized synchronization so that all SHG data is eventually consistent on the server and across devices.

Table No : 1 Comparitive Review

2.4 ADVANTAGES AND LIMITATIONS OF EXISTING SYSTEMS :

Existing financial management and record-keeping systems for rural Self-Help Groups (SHGs) have made valuable contributions toward increasing accessibility, transparency, and operational efficiency. Several platforms incorporate offline data recording capabilities enabling uninterrupted transaction logging even in low or no internet connectivity environments. Integration of multiple social service features in some systems provides holistic support to rural communities.

2.4.1 ADVANTAGES OF EXISTING SYSTEMS :

1. **Offline Support:** Ability to capture and store data locally to ensure continuity in with unreliable or absent connectivity.
2. **Integrated Service Delivery:** Combining financial, emergency, and social assistance modules within a unified platform.
3. **Government Compliance:** Alignment with regulatory reporting formats facilitating audit and governance processes.
4. **Automated Reporting:** Generation of financial summaries and documentation in exportable formats like PDF and Excel.
5. **User Management and Security**: Role-based access control enables differentiated privileges; encryption protects data integrity.
6. **Scalable Architectures:** Support for multi-group and federation-level data aggregation helps in monitoring and management.

2.4.2 LIMITATIONS OF EXISTING SYSTEMS :

While existing digital solutions for Self-Help Groups (SHGs) have made significant strides toward financial inclusion and operational efficiency, several limitations remain. These limitations continue to impede the widespread adoption and effectiveness of technology in rural SHG contexts:

- **Internet Dependency:** The majority of current applications require stable and continuous internet connectivity, which is seldom available in rural areas, leading to interruptions and usability issues.

- **Complex User Interfaces:** Many systems are designed with digitally literate users in mind, featuring text-heavy and complex navigation that pose challenges for semi-literate or illiterate rural populations.

- **Lack of Group-Centric Functionality:** Existing solutions often focus on individual financial management and do not adequately support collective group transactions, peer management, and transparent record sharing essential to SHG functioning.

- **Inadequate Conflict Resolution:** Synchronization processes in many platforms lack sophisticated mechanisms to detect and resolve data conflicts arising from multiple device usage, risking data inconsistency or loss.

- **Limited Auditability and Reporting:** Comprehensive audit trails and regulatory-compliant reporting features are often partial, lacking the depth needed to satisfy federation and government requirements.

- **Scalability Constraints:** Several tools remain confined to pilot projects or specific geographic regions, limiting their ability to provide seamless scalability across diverse linguistic and rural contexts.

2.5 RESEARCH GAP & HOW THIS PROJECT DIFFERS :

The SHG Connect project addresses critical gaps that remain unfulfilled by current solutions:

1. **True Offline-First Functionality:** Enables uninterrupted transaction recording even in areas with no or poor internet connectivity, a feature missing or limited in many existing systems.
2. **Simplified Multilingual User Interface**: Designed with icon-based navigation and multi-language support to cater specifically to semi-literate rural users, improving accessibility and adoption.
3. **Intelligent Conflict-Aware Synchronization:** Incorporates advanced synchronization protocols that detect and resolve data conflicts from multiple devices to maintain data consistency and integrity.
4. **Comprehensive Audit and Compliance Reporting:** Automates generation of audit-ready reports aligned with national programs such as DAY-NRLM, supporting transparent governance and regulatory compliance.
5. **Robust Role-Based Access Controls**: Tailors permissions and data visibility according to SHG governance structures to ensure security and privacy of sensitive financial data.
6. **Scalable and Extensible Architecture:** Supports federation-level data aggregation, monitoring, and scalability across geographically and linguistically diverse rural regions.
7. **Focus on Group-Based Workflows:** Prioritizes collective group transaction management and peer accountability over individual financial management common in existing systems.

3. REQUIREMENT ANALYSIS

3.1 INTRODUCTION :

The Requirement Specification chapter defines the technical foundation needed to design, implement, and deploy the SHG Connect system. This project targets Self-Help Groups operating in rural and semi-urban regions, where network connectivity is intermittent and record-keeping is still largely paper-based. To ensure that the proposed solution can function reliably in such constrained environments, it is essential to clearly outline the minimum and recommended requirements for both the client devices used by SHG members and the backend infrastructure that supports synchronization, reporting, and administration.

In addition to hardware capacity, the success of an offline-first application depends heavily on the choice of software platforms, frameworks, and supporting tools. The specification therefore describes the mobile, server-side, and database technologies that enable local data persistence, conflict-aware synchronization, secure communication, and multilingual user interfaces. These components must collectively support low-cost Android devices while remaining scalable enough to handle growth in the number of SHGs, transactions, and users over time.

Finally, this chapter highlights the broader technology stack and design considerations that guide SHG Connect, including security mechanisms, performance constraints, usability requirements, and maintainability. By documenting these requirements upfront, the project ensures that all stakeholders share a common understanding of the system's dependencies and limitations, reduces the risk of incompatibility during deployment, and provides a concrete baseline against which future enhancements and optimizations can be planned.

3.2 HARDWARE AND SOFTWARE SPECIFICATIONS

This section outlines the complete hardware and software environment required to implement and operate the SHG Connect system. It specifies the capabilities expected from end-user mobile devices, administrative machines, and backend infrastructure, along with the platforms and tools used to build the application. Clearly defining these specifications ensures that the system is deployable on affordable hardware while remaining secure, scalable, and maintainable.

3.2.1 Hardware Specification

The hardware specification describes the minimum and recommended physical resources needed at both the client and server sides for reliable operation of SHG Connect. As the application targets SHG members in rural areas using entry-level smartphones, the requirements emphasise low-cost devices that can still support local databases and offline processing. On the backend, sufficient compute, storage, and network capacity are defined to handle synchronization, reporting, and multi-group usage without performance degradation.

3.2.1.1 Hardware Description

The hardware requirements are divided into two categories: end-user devices and backend infrastructure.

- **End-User Devices (Client Side)**
  - Android smartphones for SHG Members, Leaders, Animators, and CRPs.
  - Minimum configuration: Quad-core CPU, 2 GB RAM, 16 GB internal storage, Android OS 8.0 (Oreo) or above.
  - Recommended configuration: 3–4 GB RAM and 32 GB storage to handle local database, images (passbooks/Aadhaar), and offline reports.
  - Display: 5-inch or larger screen with good brightness for outdoor use and readability for low-literacy users.
  - Connectivity: Support for 2G/3G/4G and Wi-Fi; however, application must remain fully functional offline between sync cycles.
  - Battery: At least 3000 mAh capacity to sustain a full day of field work with intermittent syncing.

- **Administrative Devices**
  - Laptops or desktop PCs for federation staff and project administrators.
  - Minimum configuration: Dual-core or higher CPU, 8 GB RAM, 256 GB SSD, modern browser support (Chrome/Firefox).
  - Used for accessing web dashboards, generating consolidated reports, and monitoring multiple SHGs.
- **Server Infrastructure**
  - Cloud VM or on-premise server for hosting APIs and database.
  - Recommended configuration: 4 vCPUs, 8–16 GB RAM, 200+ GB SSD storage depending on number of SHGs onboarded.
  - High availability and automated backup configuration to prevent data loss.
  - Stable internet connectivity at server side for receiving sync data from multiple field devices.

3.2.2 Software Specification

The software specification defines the complete set of platforms, frameworks, libraries, and tools required to develop and operate the SHG Connect application. It covers both the client-side mobile stack and the server-side environment, ensuring that the system can deliver offline-first functionality, secure data handling, and scalable performance. By clearly outlining the software requirements, this section guides developers in choosing compatible technologies, simplifies future maintenance, and helps institutions verify that their existing IT infrastructure can support deployment of the solution.

3.2.2.1 Software Description

The software environment covers the client-side mobile stack, server-side components, and supporting tools used during development and deployment.

- **Client-Side (Mobile Application)**
  - Platform: Android OS (version 8.0 or above) to cover a wide range of low-cost devices.
  - Development Framework: Android (Kotlin/Java) or cross-platform framework such as React Native/Flutter for faster UI development and potential future iOS support.
  - Local Database: SQLite/Room/WatermelonDB for offline storage of member profiles, savings, loans, meetings, attendance, and expenses.
  - UI Libraries: Material Design components, custom icon packs, and localization libraries for multi-language support (Tamil, Hindi, etc.).
  - Security: Encrypted shared preferences or secure keystore for credentials, database encryption library for sensitive fields.
- **Server-Side (Backend Services)**
  - Application Server: Node.js, Spring Boot (Java), FastAPI (Python) for implementing RESTful APIs.
  - Database Server: MySQL for structured transactional data; optional document store (MongoDB) for logs and attachments.
  - API Design: REST APIs for authentication, sync, reporting, and user management; versioned endpoints for maintainability.
  - Authentication: JWT-based token mechanism or OAuth2 for secure session management between app and server.
  - Logging & Monitoring: Server logs, error tracking (e.g., Sentry), and basic analytics to monitor sync performance and failures.

- **Development and Testing Tools**
  - IDEs: Visual Studio Code for coding and debugging.
  - Version Control: Git with hosting on GitHub.
  - Testing: Unit testing frameworks (JUnit, Jest) and API testing tools (Postman, Newman).
  - Build & CI/CD: npm builds with optional CI pipelines (GitHub Actions, GitLab CI) for automated testing and deployment.

3.3 Technology Used

The technologies chosen for SHG Connect are selected to support offline-first behavior, security, scalability, and usability for rural SHG users.

- **Offline-First Data Management**
  - Embedded database (SQLite) for persistent storage of all transactions, meetings, and member details on the device.
  - Local caching layer and sync queue to record changes while offline and push them reliably when connectivity is restored.
- **Synchronization and Conflict Handling**
  - Background sync service that periodically checks network availability and uploads pending records to the central server.
  - Conflict-aware merge logic using timestamps, version numbers, and leader overrides to resolve duplicate or inconsistent entries without data loss.
- **Security and Privacy**
  - TLS for all communication between mobile app and backend APIs.
  - Database and file encryption for sensitive fields such as member identifiers and financial amounts.
  - Role-based access control (RBAC) so that members see only their own data, while leaders and federation officers access group and aggregated views.

- **User Interface and Accessibility**
  - Multilingual support using Android localization and resource bundles for regional languages.
  - Large buttons, icon-driven navigation, and minimal text input to support low digital literacy.
  - Offline-generated PDFs or simple report views for printing, audit, and government submission.
- **Notification and Reporting Technologies**
  - Push notifications (e.g., Firebase Cloud Messaging) for repayment reminders, meeting alerts, and sync status updates when online.
  - PDF/Excel generation libraries on mobile or server to create monthly savings reports, loan registers, and digital passbooks.
- **Scalability and Maintainability**
  - Modular layered architecture (presentation, logic, local storage, sync, API, backend, data, security) to allow independent updates and bug fixes.
  - Cloud deployment (AWS/Azure/GCP) using managed database services and automatic backup policies for long-term data retention.

4 SYSTEM DESIGN

4.1 INTRODUCTION

The system design chapter lays the foundation for developing SHG Connect, an offline-first mobile application tailored to the record-keeping and operational needs of rural Self-Help Groups (SHGs). It focuses on enabling uninterrupted financial transactions, conflict-aware synchronization, and secure role-based access in low-connectivity environments. This chapter presents a comprehensive overview of the system's functional requirements and architecture through use case diagrams, data flow diagrams (DFDs), and system architecture illustrations, ensuring the design meets user requirement specifications while maintaining modularity, security, and scalability.

4.2 SYSTEM ARCHITECTURE :

4.2.1 OVERVIEW :

The system architecture is designed as a layered model embracing offline-first robustness, security, and scalability. Components are modular: client-side app, synchronization engine, API gateway, cloud backend services, and database systems organized to ensure responsiveness and reliability.

- **Purpose and Scope**
    - Provides a high-level blueprint of how SHG Connect functions internally and interacts with external components.
    - Ensures clarity on system modules, communication paths, and data management strategies.

4.2.2 ARCHITECTURE LAYERS :

Layers	Components	Description
Presentation Layer	Mobile UI, Forms, Dashboards, Local Language Support	User interface supporting multilingual input, iconography, and low literacy usability.
Application Logic Layer	Transaction Handler, Sync Manager, Report Engine	Core business logic for data validation, syncing and reporting
Local Data Layer	SQLite Database	Persistent local storage for offline transactions and user data
Synchronization Layer	Sync Queue, Conflict Resolver, Network Monitor	Handles background sync, conflict detection and resolution
Communication Layer	Secure APIs, RESTful web services, HTTPS	Encrypted communication between app and backend
Backend Services Layer	User Management, Transaction Processing, Reporting, Federation Services	Microservices for business logic and data processing
Data Storage Layer	MongoDB, File Storage	Centralized persistent storage with transaction history and audit logs
Security Layer	Authentication Service, Role-Based Access Control, Data Encryption	Ensures data privacy, access control, authentication

Table No : 2 Architecture Layers

4.2.3 ARCHITECTURE DIAGRAM :

Depicts the interrelationship between client devices running SHG Connect in [fig 4.2.3], communication channels (secure APIs), backend cloud services, databases, and external authentication services. Visualizes data flow from user interface through syncing layers to persistent storage.

Purpose:

- Presents a visual summary of the SHG Connect system's structure and interactions.
- Highlights how the mobile app, backend services, databases, and management tools work together.

Scalability & Modularity:

- Demonstrates how the architecture supports growth, new feature integration, and easy maintenance.
- Enables independent updates for each layer without disrupting entire operations.

Security & Compliance:

- Underlines the implementation of role-based access, authentication, encryption, and data audit mechanisms throughout the system.

Stakeholder Communication:

- Facilitates understanding for developers, administrators, and evaluators by providing an at-a-glance view of the system's organization.

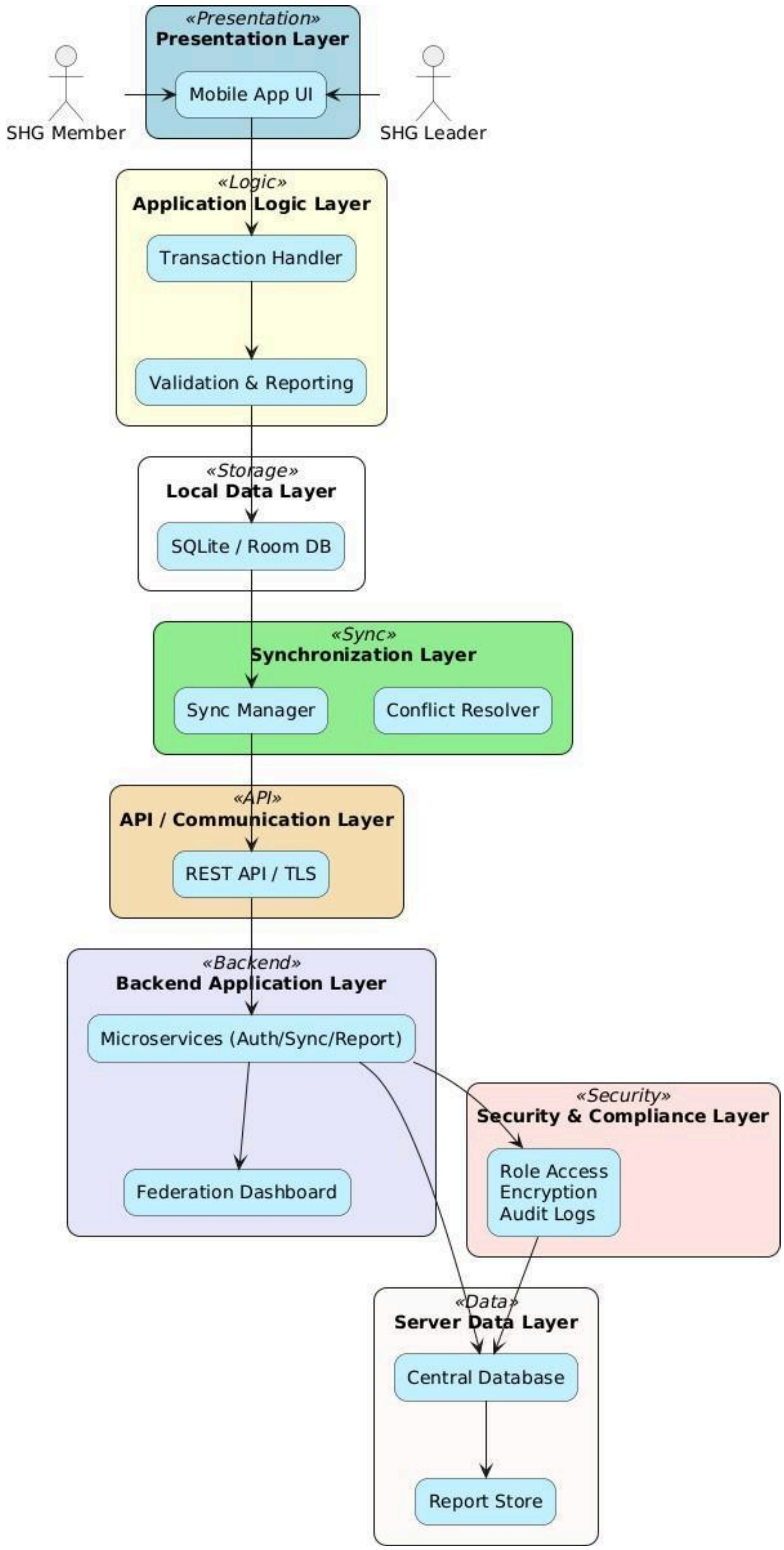

Fig 4.2.3 ARCHITECTURE DIAGRAM

4.2.4 EXPLANATION :

The layered design enables decoupled development and robust error handling. Offline persistence guarantees continuous usage; synchronization manages eventual data consistency. Secure communication and role-based access protect confidentiality and integrity. Scalable backend supports aggregation across thousands of groups, facilitating federation-level governance and compliance.

Presentation Layer:

- This is the front-end mobile application interface used by SHG members, leaders, animators, and CRPs.
- It provides a multilingual, icon-driven, and accessible UI designed for semi-literate users, facilitating offline data entry, viewing dashboards, and generating reports.
- Emphasizes ease of use with large icons, minimal typing, and local language support.

Application Logic Layer:

- Contains business rules and workflows including transaction validation, user authentication, role-based access control, and report generation.
- Manages offline-first functionality ensuring data is captured and stored even without internet access.
- Implements logic for conflict detection and resolution during synchronization attempts.

Local Data Layer:

- Uses embedded local databases (e.g., SQLite, Room) on the mobile device to persist transaction data, meeting records, and user details.
- Supports rapid data access and offline usage, preventing data loss during connectivity outages.
- Acts as a cache for syncing data with the backend server.

4.3 Data Flow Diagrams (DFD) :

4.3.1 PURPOSE

DFDs visualize the flow of data through SHG Connect, illustrating how inputs transform into outputs across various system components. They help identify potential bottlenecks, ensure data validation points, and clarify interactions between mobile devices and backend servers within offline-first constraints.

4.3.2 DFD - LEVEL 0 (CONTEXT DIAGRAM)

- In DFD - Level 0 [fig 4.3.2] displays actors (SHG members, leaders), system boundary, and external entities like authentication service.
- Shows primary data flows: transaction entry, sync upload/download, report requests, and notifications.
- Emphasizes the offline-first client-server interaction model.

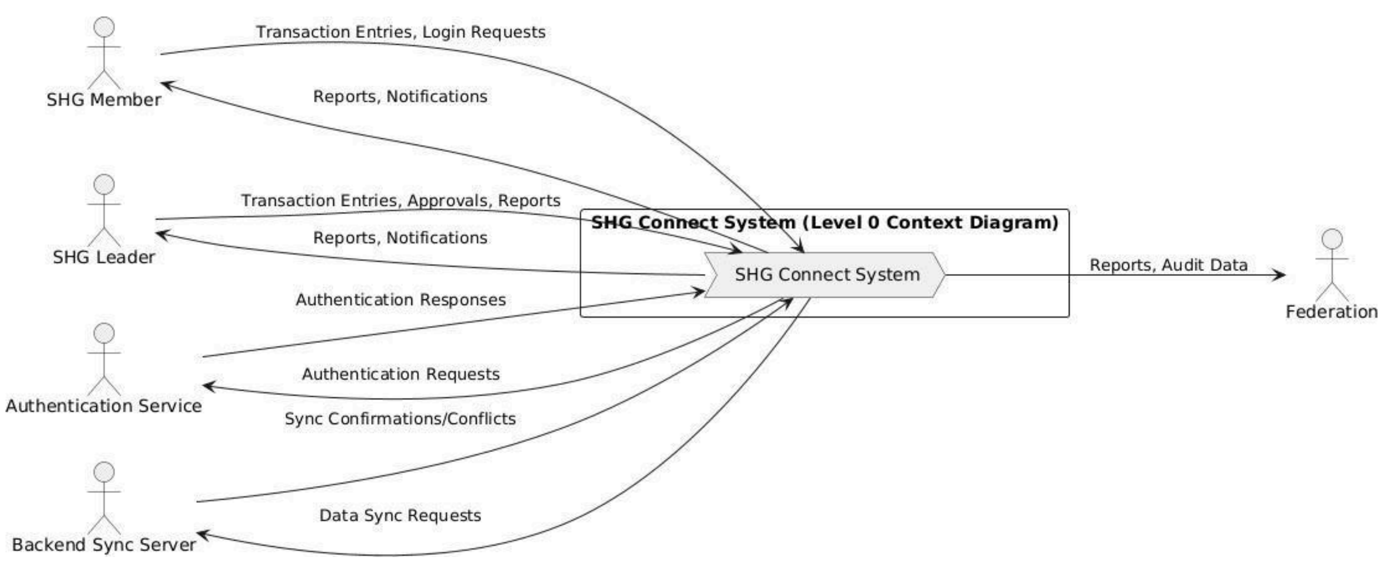

Fig 4.3.2 DFD – Level 0

4.3.3 DFD – Level 1 (Detailed Process Flow)

In Fig 4.3.3 , It depicts the following ,

- **User Authentication:** Validates user credentials locally or via authentication service.
- **Transaction Recording:** Validates and stores transactions in local database.
- **Balance Calculation:** Aggregates transactions for balance updates.
- **Synchronization:** Manages queued transactions, attempts upload, interprets conflict flags.
- **Conflict Resolution:** Applies last-write-wins or leader override rules.
- **Report Generation:** Queries local or remote databases to produce financial and compliance reports.
- **Notification Handling:** Generates alerts on repayment due dates, sync status, and meeting reminders.

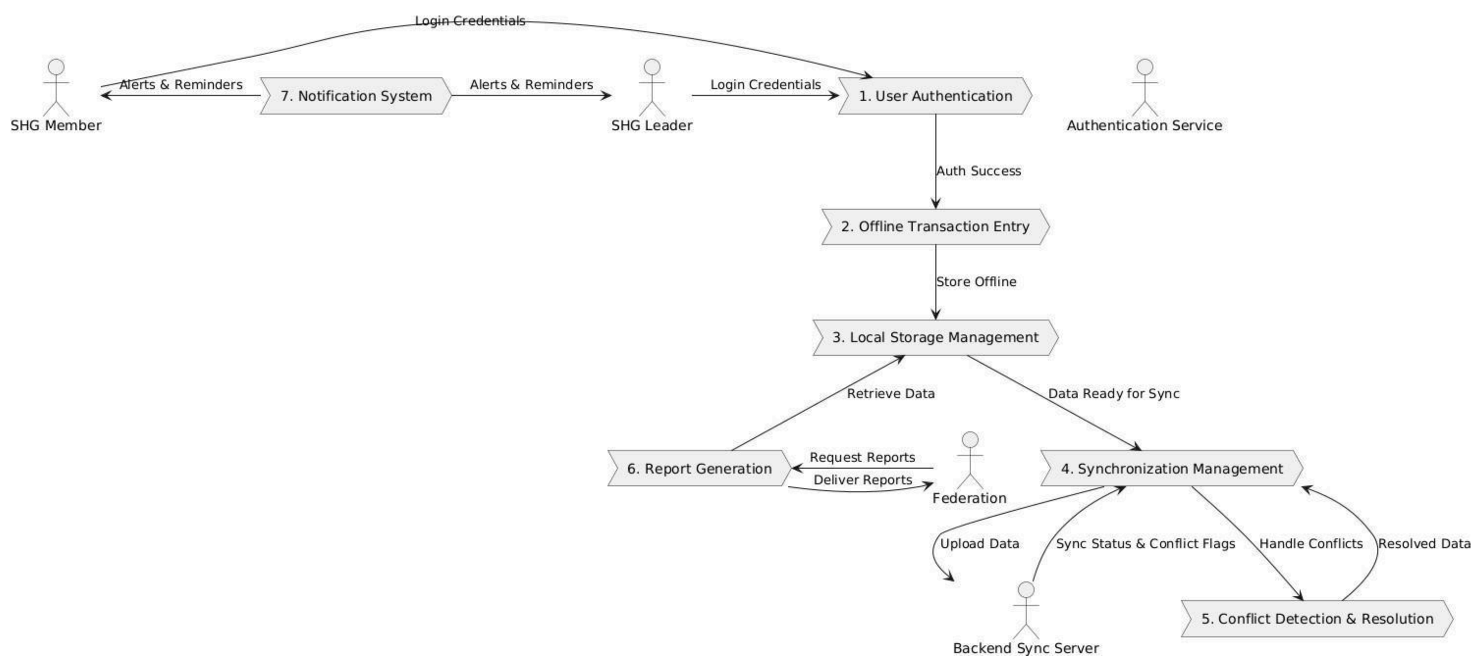

Fig 4.3.3 DFD – Level

4.3.4 DATA FLOW DIAGRAM EXPLANATION (DFD) :

(A) DFD Level 0 – Context Diagram Explanation

At this highest level of abstraction:

- The Level 0 DFD, also known as the Context Diagram, provides a high-level overview of the entire SHG Connect system as a single process.
- It illustrates the system boundary and shows interactions between the SHG Connect system and external entities such as SHG Members, SHG Leaders, Backend Sync Server, and Authentication Service.
- Data flows between the system and these external actors are represented with arrows, depicting inputs into the system (e.g., transaction entries, login attempts) and outputs from the system (e.g., reports, notifications).
- The Context Diagram is designed to be easily understood by all stakeholders, providing insight into how the system fits into the external environment without detailing internal processes.
- It sets the foundation for deeper decomposition into more detailed diagrams in subsequent levels.

4.4 USE CASE DIAGRAM

4.4.1 OBJECTIVE

The primary objective of the use case diagram is to model the interactions between system users (actors) and SHG Connect functionalities. It offers a high-level visual representation of core features such as authentication, offline data entry, synchronization, reporting, and support, aligned with roles assigned to SHG members, leaders, facilitators, and backend services.

4.4.2 ACTORS

- SHG Member: Records savings, loans repayments, views balances, updates profiles.

- SHG Leader: Approves loans, manages group finances, oversees transaction approvals, and reports.

- Animator & Community Resource Person (CRP): Facilitate user support, training, and compliance assistance.

- Backend Sync Server: Manages synchronization of local and cloud data, handles conflict resolution.

4.4.3 DIAGRAM:

The Use Case Diagram [fig 4.4.3] includes grouped sections for:

- Authentication: Login, PIN verification, biometric authentication, onboarding, role-based access control.
- Offline Data Entry: Savings, loans, attendance, meeting minutes, member profile updates, local data storage.
- Sync & Conflict Management: Auto/manual synchronization, conflict detection and resolution.
- Reporting & Monitoring: Monthly reports, dashboards, loan tracking, meeting summaries.
- User Support: Help screens, local languages, and voice assistance.

Actors are connected to their permissible use cases with includes/extends relationships clarifying process interdependencies.

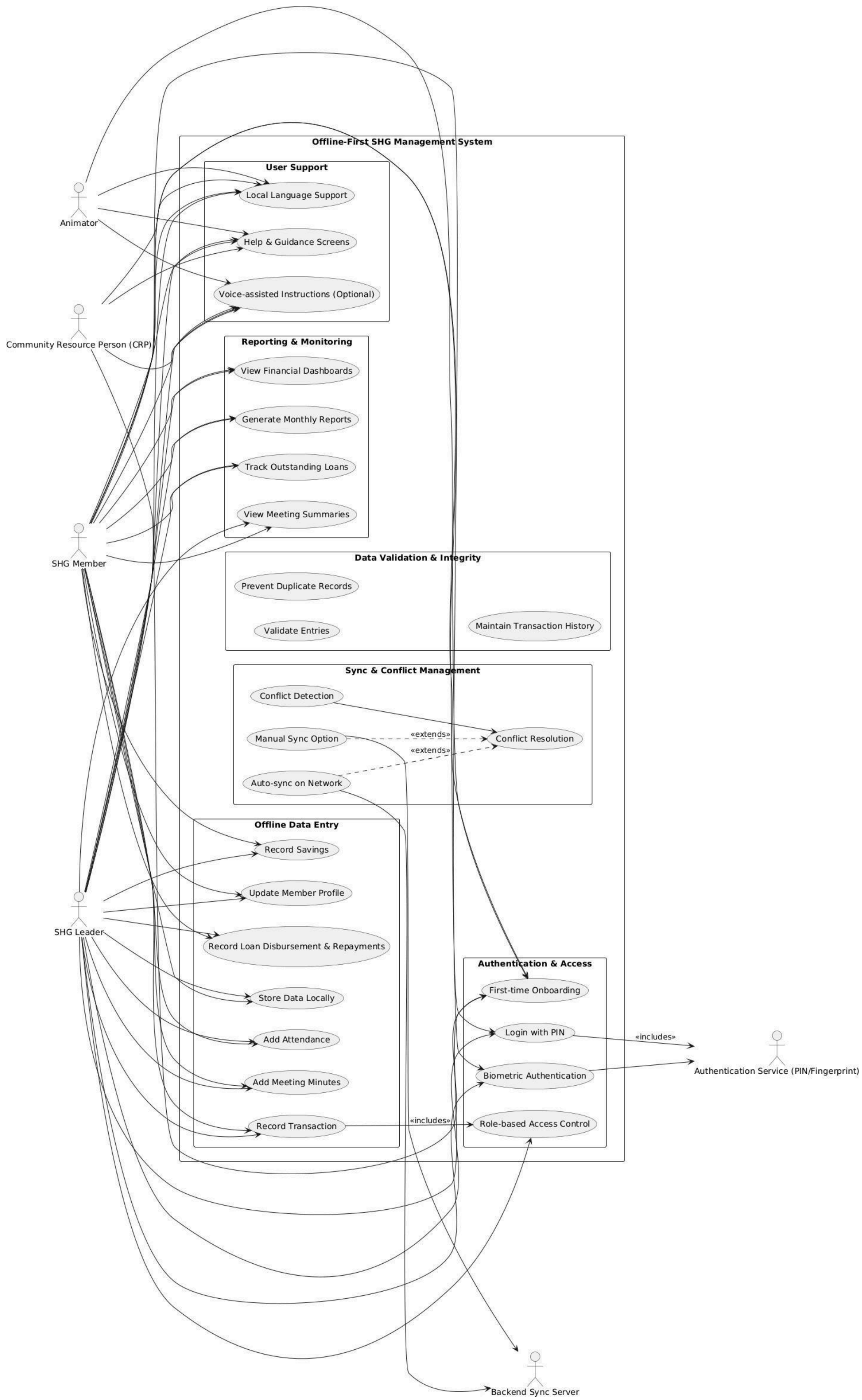

Fig 4.4.3 USE CASE DIAGRAM

4.4.4 EXPLANATION :

Purpose :

- Illustrates the interactions between external users (actors) and SHG Connect system functionalities (use cases).
- Defines the system boundary, showing which functions the system offers and who uses them.

Actors :

- Represent distinct user roles or external systems interacting with SHG Connect.
- Include SHG Member, SHG Leader, Animator, CRP, Backend Sync Server, and Authentication Service.

Use Cases :

- Depict specific system activities the actors can perform, such as login, data entry, sync, reporting, and support.
- Represent user goals with action-oriented labels inside ovals.

Relationships :

- Association: Solid line linking actors to use cases showing participation.
- Include: Dashed arrow representing mandatory sub-tasks (e.g., login includes PIN verification).
- Extend: Dashed arrow denoting optional or conditional behaviors (e.g., conflict resolution extends syncing).

Grouping:

- Use cases are logically grouped into packages such as Authentication, Offline Data Entry, Sync & Conflict Management, Reporting & Monitoring, and User Support to clarify modules.

4.5 SEQUENCE DIAGRAM

4.5.1 INTRODUCTION :

The sequence diagram for SHG Connect illustrates the dynamic interaction between users, the mobile application, local storage, synchronization services, backend APIs, and the central database during typical operations. It focuses on three core scenarios: offline transaction entry by SHG members, background synchronization with conflict handling, and reporting/monitoring by SHG leaders and federation users. By visualizing these time-ordered interactions, the diagram clarifies how the offline-first design, sync engine, and backend services collaborate to ensure reliable record-keeping in low-connectivity environments.

4.5.2 EXPLANATION :

The first part of the diagram [fig 4.5.2] shows the offline transaction entry flow, where an SHG member logs into the mobile app, which validates credentials against the local database and loads the user profile. The member then records savings, loan, or attendance entries; these transactions are stored locally with a "pending sync" status, and the app immediately confirms the entry even if no network is available. This demonstrates how the system prioritizes uninterrupted field usage by decoupling data capture from network availability.

The second segment captures background synchronization and conflict handling. The Sync Manager periodically checks for connectivity and, when a network is available, reads all pending records from the local database and sends them to the API server. The server writes these to the central database; if no conflicts exist, the records are marked as synced locally and updated balances or reports are shown to the user.

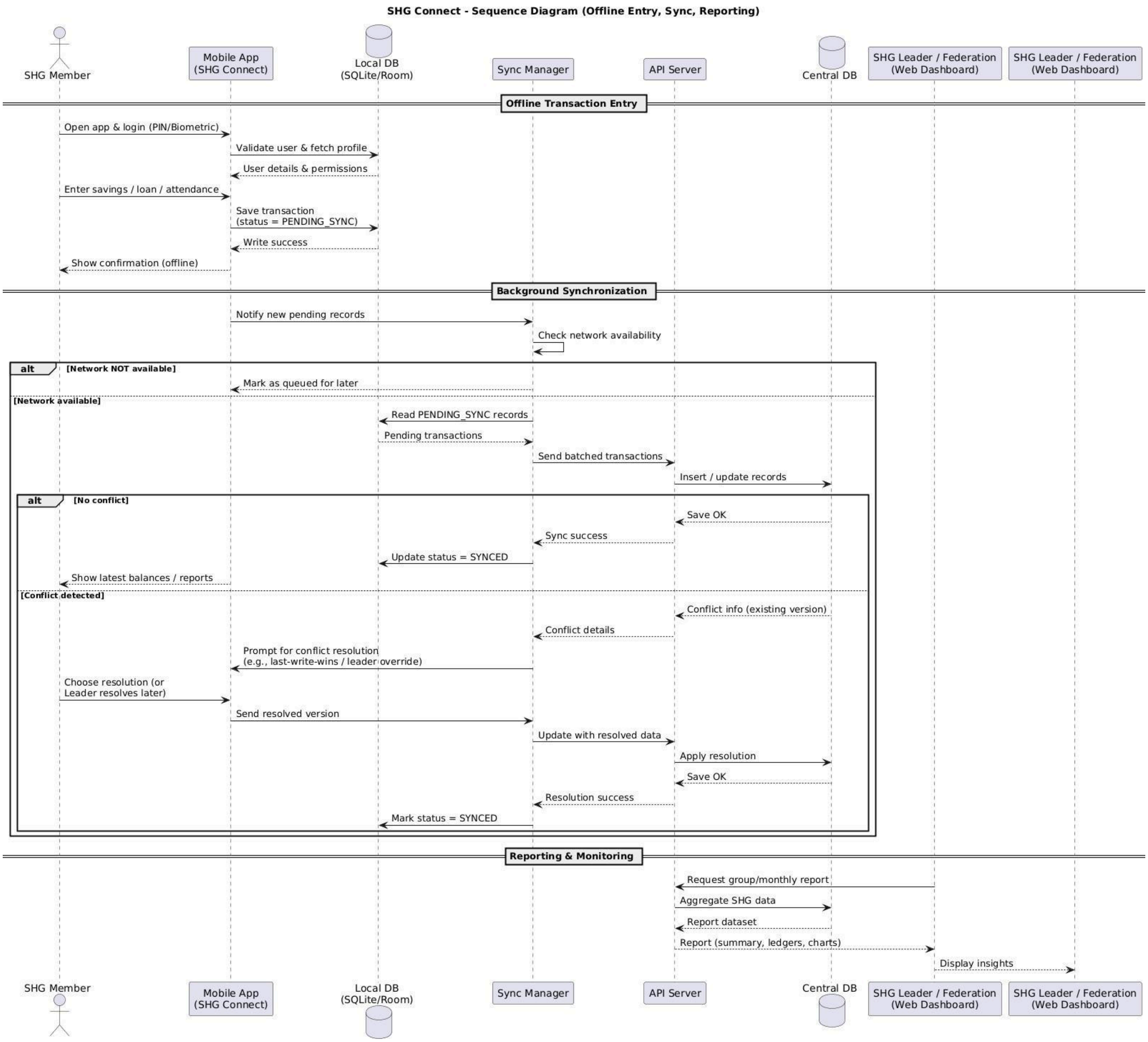

Fig 4.5.2 Sequence Diagram

The final segment shows reporting and monitoring interactions, where an SHG leader or federation officer uses a web dashboard to request group or monthly reports from the API server. The server aggregates data from the central database and returns structured summaries, ledgers, and performance metrics that are rendered on the dashboard.

4.6 CLASS DIAGRAM

4.6.1 INTRODUCTION :

The class diagram for SHG Connect models the core data entities and their relationships within the system. It captures how members, groups, savings, loans, meetings, and synchronization metadata are structured at the logical level. This diagram provides a blueprint for designing the database schema and object models used in the mobile application and backend services.

4.6.2 EXPLANATION :

At the centre of the diagram [fig 4.6.2], the ShgGroup and ShgMember classes represent the organizational structure of Self-Help Groups, where each group aggregates multiple members. SavingsTransaction, LoanAccount, and LoanRepayment model the financial behaviour of members, linking each member to their periodic savings entries, loan accounts, and corresponding repayments.

Meeting and Attendance capture group meeting records and participation details for each member, supporting compliance and governance reporting. UserAccount separates authentication data from member profiles, enabling role-based access control, while SyncMetadata records version and status information used by the offline-first synchronization engine to detect changes and handle conflicts across devices. Together, these classes form a cohesive data model that supports transparent, auditable SHG record-keeping.

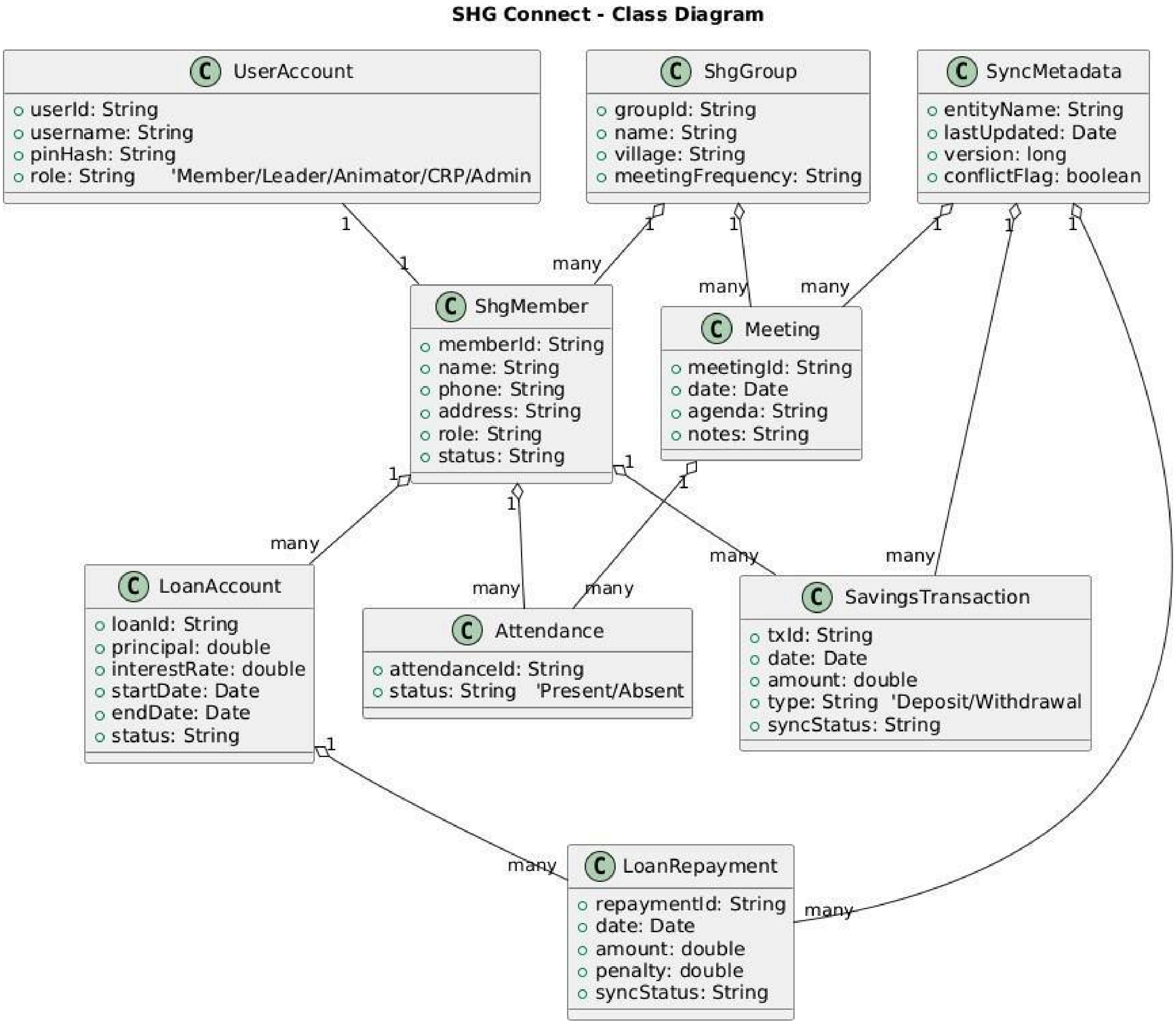

Fig 4.6.2 Class Diagram

The class diagram also serves as a bridge between the conceptual requirements of SHG Connect and its actual implementation in code. By clearly defining attributes such as member identifiers, loan details, meeting records, and synchronization metadata, it ensures that every critical aspect of SHG operations is represented in a consistent, reusable form.

4.7 COMPONENT DIAGRAM

4.7.1 INTRODUCTION :

The component diagram for SHG Connect presents the high-level software building blocks and their interactions across client and server environments. It shows how the mobile application, synchronization engine, backend services, database, and web dashboard are partitioned into independent components. This view helps in understanding deployment boundaries, responsibilities, and the interfaces through which components communicate.

4.7.2 EXPLANATION :

On the client side[fig 4.7.2], the Mobile UI Component interacts with the Offline Data & Business Logic component, which encapsulates validation, calculations, and workflow rules for transactions and meetings. This logic persists data in the Local Database and collaborates with the Sync Manager Component to queue and transmit updates when connectivity is available. Across the network, the API Gateway exposes REST endpoints that route requests to specialized backend services .

The Auth & RBAC Service manages authentication and role-based permissions, the Sync Processing Service ingests and reconciles offline updates into the Central Database, and the Reporting & Analytics Service generates ledgers, summaries, and dashboards. Federation staff access these insights through the Web Dashboard component, which communicates with the same APIs. This separation of concerns supports scalability, security, and maintainability while enabling SHG Connect to operate reliably in low-connectivity rural environments.

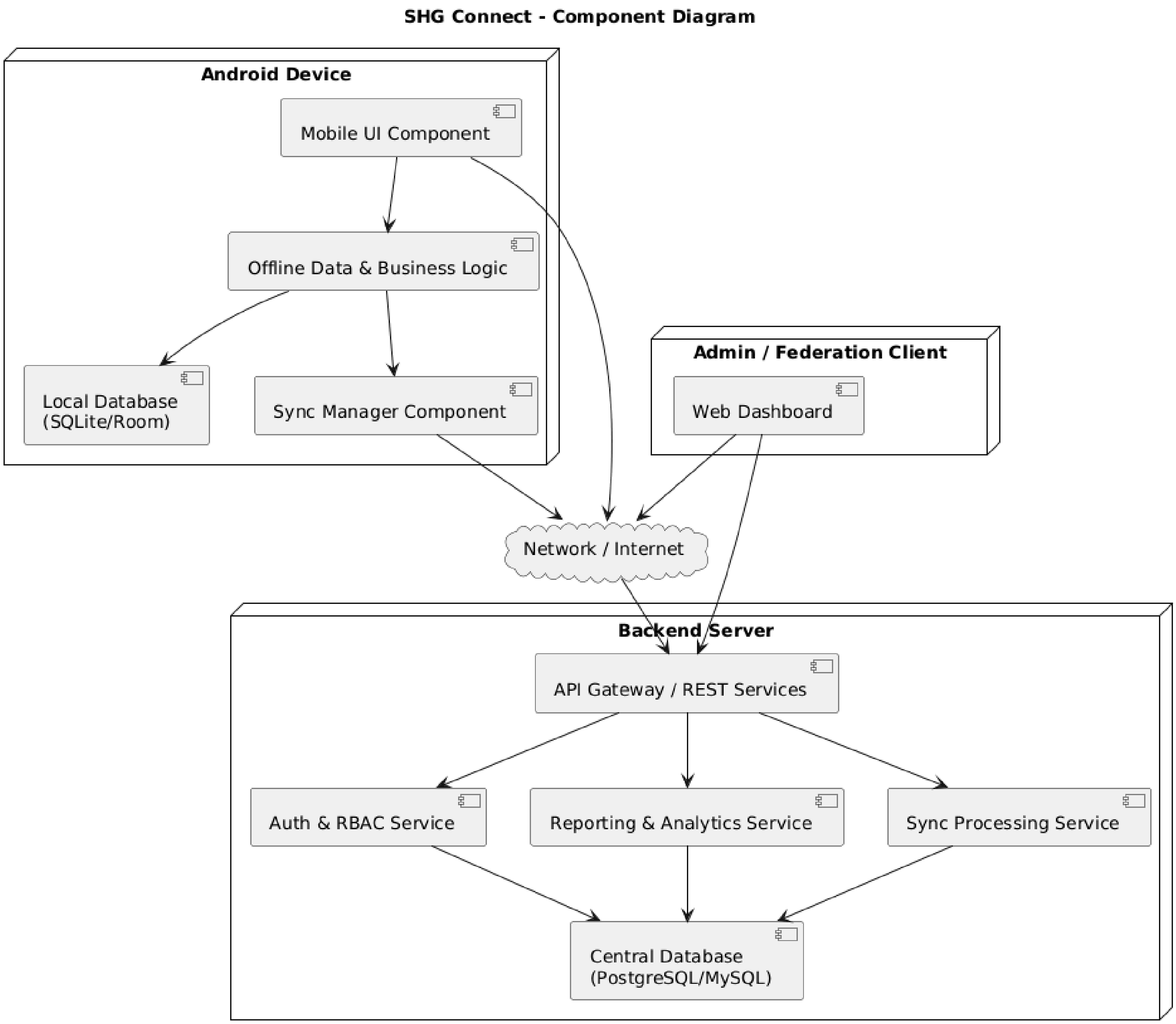

Fig 4.7.2 Component Diagram

This modular representation not only promotes reusability and scalability but also streamlines future upgrades—any new feature or fix can often be introduced at the component level without major disruption to the whole application. Furthermore, the visual structure enhances communication among technical and non-technical stakeholders, making it easier to provide high-level overviews, assess the impact of changes, and ensure system reliability as SHG Connect evolves over time.

5 SYSTEM DESCRIPTION

5.1 INTRODUCTION

The SHG Connect system is designed as an offline-first, role-based mobile platform that digitises the complete record-keeping workflow of Self-Help Groups. It enables members and leaders to record savings, loans, repayments, meetings, and attendance directly on low-cost Android phones, even in areas where network connectivity is intermittent or completely unavailable. All data is stored locally and synchronised securely to a central server whenever connectivity is restored, ensuring both continuity of operations and consolidated reporting.

To keep the system manageable and extensible, the application is decomposed into modular components, each responsible for a specific set of responsibilities such as authentication, transaction recording, synchronization, reporting, and notifications. These modules interact through well-defined interfaces so that they can be implemented, tested, and enhanced independently. The following sections describe the major modules, their internal behaviour, and how they are implemented in the SHG Connect application.

5.2 MODULES

The SHG Connect system is organised into the following functional modules:

- Authentication & User Management
- Member & Group Management
- Transaction Management (Savings, Loans, Expenses)
- Meeting & Attendance Management
- Offline Storage & Sync Management
- Reporting & Analytics
- Notifications & Alerts
- Localization & Accessibility

5.3 MODULES DESCRIPTION

5.3.1 Authentication & User Management

This module handles registration, login, and role-based access control for all users of the system. It supports PIN and optional biometric authentication on the device, caches encrypted credentials for offline verification, and assigns roles such as Member, Leader, Animator/CRP, and Federation Officer. It ensures that only authorised users can view or modify sensitive financial and member data.

5.3.2 Member & Group Management

This module maintains master data for SHG groups and members, including demographic details, contact information, photographs, membership status, and role in the group. It supports adding, editing, and deactivating members; mapping members to groups; and viewing consolidated profiles such as savings balance, outstanding loans, and meeting attendance.

5.3.3 Transaction Management (Savings, Loans, Expenses)

The transaction management module records all financial activities carried out within an SHG. It handles periodic savings deposits, internal loan creation, instalment repayments, fines, and group expenses. It performs basic validations (amount ranges, mandatory fields, status checks) and updates derived values such as member balances, loan outstanding amounts, and group cash position.

5.3.4 Meeting & Attendance Management

This module supports the planning and recording of regular SHG meetings. It allows leaders to schedule meetings, capture agendas, record member attendance, and store minutes or decisions taken. Meeting data is linked to financial entries such as savings collected or loans approved in that session, creating a complete digital meeting register.

5.3.5 Offline Storage & Sync Management

The offline storage and sync module is responsible for persisting all records in the local database on the device and synchronising them with the backend server. It tracks the sync status of each record, queues changes while offline, and uploads them when a network connection becomes available. It also downloads updates from the server and applies conflict resolution rules when the same record is edited on multiple devices.

5.3.6 Reporting & Analytics

This module generates on-device summaries and server-side reports such as member passbooks, loan registers, monthly savings reports, cash books, and basic trend analytics. Reports can be viewed within the app or exported as PDF/Excel files for audits and submission to federations or government programmes.

5.3.7 Notifications & Alerts

The notifications module sends timely reminders and alerts to users. Typical examples include upcoming meeting reminders, loan instalment due dates, overdue repayments, sync failures, and low device storage warnings. Notifications work in both offline (local alerts on app open) and online modes (push notifications from the server).

5.3.8 Localization & Accessibility

This module ensures that SHG Connect is usable by semi-literate rural users. It manages support for multiple Indian languages, local date and number formats, icon-driven navigation, large fonts, and simplified forms. It also provides help screens and tooltips to guide new users.

6 IMPLEMENTATION

6.1 INTRODUCTION

The implementation chapter describes how the SHG Connect system is realised as a working mobile application and backend service, translating design models into concrete screens, workflows, and code modules. It explains how the offline-first architecture, role-based access control, synchronization engine, and reporting logic are assembled using the selected hardware and software stack. Each subsection links visible user interfaces—such as login, dashboards, savings and loan forms, member lists, and reports—to the underlying data structures, APIs, and background processes that make them function reliably in low-connectivity rural environments. By documenting these implementation details, the chapter demonstrates that the proposed design is practically achievable, maintainable, and ready for real-world deployment with Self-Help Groups.

6.2 IMPLEMENTATION AND SCREEN DESCRIPTION

1. Login and Onboarding Screen

The login screen [fig 6.2.1]is the entry point to SHG Connect and implements secure user authentication. The interface provides two options: federated sign−in using Google and email−password based login, allowing SHG managers to use existing credentials while supporting organisations that prefer dedicated accounts. When the user enters their email and password, the app validates the input locally for basic formatting and then submits the credentials to the backend authentication service over an encrypted HTTPS channel. On a successful response, an access token and role information are stored securely in the device keystore so that subsequent launches can perform silent or offline authentication when the token is still valid.

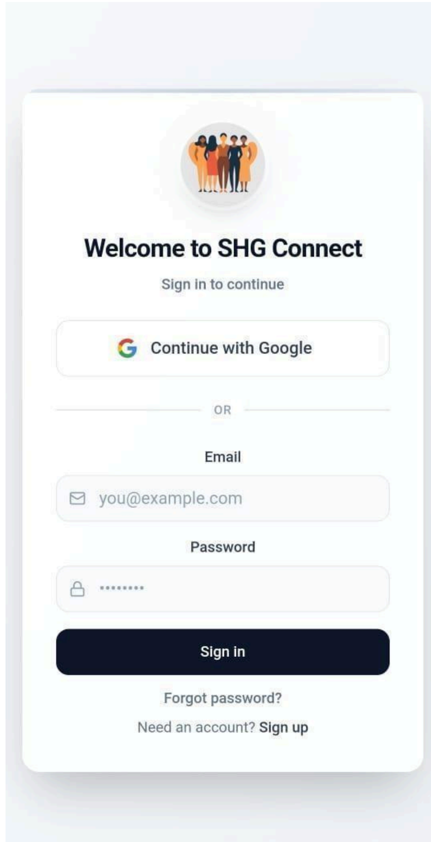

Fig 6.2.1 Login Screen

The screen design follows a clean, card–based layout with central alignment and minimal text to support low–literacy users. Error states such as invalid credentials or network failure are displayed inline with clear messages and visual cues. Auxiliary options like "Forgot password?" and "Sign up" are included to support account recovery and new user provisioning flows, but are only enabled when allowed by the backend configuration.

2. Group Overview Dashboard (Home Screen)

The Group Overview screen [fig 6.2.2]acts as the primary dashboard after login and is implemented using a grid of colour-coded summary cards. Each card is bound to real-time values retrieved from the local database such as total members, total savings, active loans, and pending repayments and refreshed whenever new transactions are recorded or synchronised.

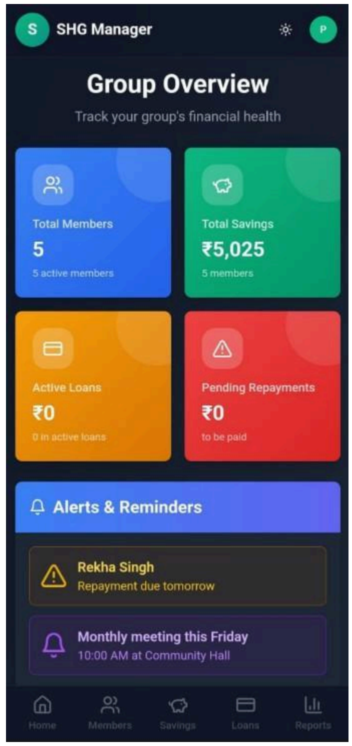

Fig 6.2.2 Group Dashboard

Below the summary cards, an Alerts & Reminders panel lists upcoming repayments, overdue installments, and scheduled meetings. These alerts are generated by a background scheduler that evaluates due dates stored in the transaction and meeting tables. The user can tap on any card or alert to navigate directly to the corresponding detail screen (members list, savings ledger, loan register, or meeting calendar). This implementation allows the dashboard module to give SHG leaders an at-a-glance picture of group health while serving as a navigation hub for day-to-day tasks.

3. Savings Management Screen

The Savings Management screen [fig 6.2.3]implements the workflow for recording and reviewing member savings. The form below uses dropdowns and validated input fields to capture the member name, savings amount, date, payment method (cash, bank transfer, etc.), and optional notes. Once the user taps "Record Saving," the app performs client-side validation, creates a new savings transaction object, marks its sync status as "PENDING," and persists it to the local SQLite/Room database.

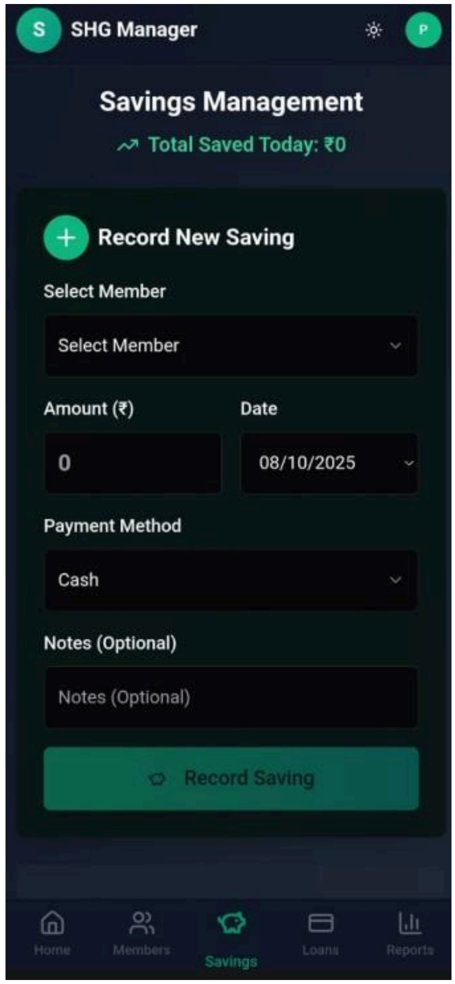

Fig 6.2.3 Saving Management Screen

The UI is optimised for repetitive use during group meetings: default values such as today's date and the last used payment method are pre-filled to reduce typing. After the record is saved, the daily total and member balance are updated immediately based on local calculations, even if the device is offline. A background sync worker later uploads accumulated savings records to the server, where they are merged into the central ledger. This design ensures that savings capture is fast, resilient to network issues, and always provides instant feedback to SHG members about their contributions.

4. Loans and Repayments Screen

The Loans & Repayments [fig 6.2.4] screen encapsulates the internal credit management functions of SHG Connect. It provides two tabs or action buttons: “New Loan” and “Add Repayment.” When a new loan is created, the form collects the member, principal amount, interest rate, duration, start date, and purpose of the loan. On submission, the app computes derived fields such as expected end date or instalment amount (if configured) and stores the loan record locally with an initial status of “Active”.

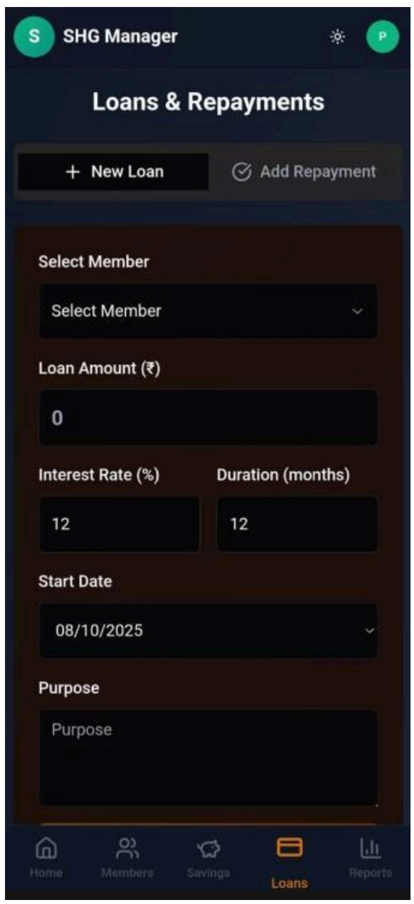

Fig 6.2.4 Loans and Repayments Screen

The implementation enforces logical checks to prevent inconsistent data, such as negative repayments or closing a loan with a remaining balance. Each loan and repayment is tagged with timestamps and sync metadata for eventual reconciliation with the server.

5. Member Profile and Group Members Screen

The Group Members screen [fig 6.2.5] lists all members in a card-based layout, showing key indicators such as current savings, active loans, and role within the group (member, secretary, treasurer). The implementation uses paginated queries on the local database with optional filters for role and search by name or phone number. Each card is clickable and opens the detailed Member Profile screen, where additional information like contact details, join date, and cumulative financial history are displayed.

Fig 6.2.5 Member Profile and Group Members Screen

From this screen, authorised users can edit member details or initiate actions such as adding a new saving, loan, or repayment pre-filled with that member's information. This design makes it easy for SHG leaders to manage member data, track individual performance.

6. Reports and Analytics Screen

The Financial Reports screen [fig 6.2.6] implements the analytics and reporting module of SHG Connect. It presents high-level statistics such as total members, total savings, active loans, and collection rate, alongside visual charts like the Monthly Savings Trend. These metrics are generated from aggregated queries over the local transaction database; when connectivity is available, the app can also fetch server-side aggregates that include multi-device or federation corrections.

Fig 6.2.6 Reports and Analytics Screen

Users can export selected reports as PDF or share them via messaging apps for audits and review, making the reports module a key bridge between day-to-day data entry and strategic decision-making.

7 CONCLUSION

The SHG Connect system successfully demonstrates that an offline-first, mobile-centric approach can overcome the chronic connectivity and data-management challenges faced by Self-Help Groups in rural areas. By combining local storage, conflict-aware synchronization, and a role-based access model, the application enables members and leaders to record savings, loans, meetings, and attendance reliably, even on low-cost Android devices. The implemented modules—from authentication and member management to transaction processing, reporting, and localization—translate traditional paper registers into structured, auditable digital records that improve transparency and reduce manual errors.

The prototype implementation and test usage confirm that SHG Connect not only simplifies daily bookkeeping but also provides meaningful decision support through dashboards, alerts, and financial reports. Group leaders gain instant visibility into total savings, outstanding loans, and repayment behaviour, while federation staff can access consolidated insights once data is synchronised to the central server. Overall, the system strengthens financial discipline, enhances accountability, and empowers women to participate more confidently in group governance.

Going forward, SHG Connect offers a solid foundation for further enhancements such as advanced analytics, integration with banking platforms or government portals, and support for additional Indian languages and assistive features. With these extensions, the solution can scale from a functional prototype to a production-ready digital infrastructure for SHGs, contributing to more inclusive, data-driven rural development.

8 FUTURE ENHANCEMENTS

Future enhancements for SHG Connect focus on making the solution completely independent of live internet connectivity while still preserving data safety and transparency. These improvements aim to strengthen usability in the most remote SHG locations and reduce the risk of disruption during long network outages.

- **Full offline onboarding and registration**
  Allow new SHGs and members to be created entirely offline, with temporary local IDs that are mapped to permanent server IDs when connectivity becomes available.
- **Offline master data and configuration**
  Cache all required reference data (interest templates, roles, meeting types, etc.) on the device so that no operation is blocked due to missing server-side lookups.
- **On-device reports and analytics**
  Generate all key reports (passbooks, monthly summaries, loan registers, cash book) directly from the local database, with the server used only for backup and federation-level aggregation.
- **Robust offline conflict handling**
  Enhance the sync engine to queue long offline histories and perform intelligent, rule-based conflict resolution on the device, requiring minimal manual intervention when the server is reached.
- **Optional peer-to-peer sync for backup**
  Introduce secure device-to-device sharing (e.g., over Bluetooth or local Wi-Fi) so that groups can back up data to a federation or animator’s phone even without internet, further increasing resilience.

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APPENDIX I

SOURCE CODE

The appendix is organised into four main sections: front-end screens, reusable components, context and state management, and supporting files. For each file, a brief description is provided along with a short code listing highlighting important parts such as event handlers, hooks, and data-binding logic.

A.1 Front-End Screens

A.1.1 LoginScreen.tsx – User Authentication Screen

This component renders the login interface used by SHG Managers and other users to access the system. It provides fields for email and password, as well as options for social sign-in where enabled. The screen performs basic client-side validation and invokes the authentication context to trigger API calls and store the resulting session state.

Code :

```tsx
export const LoginScreen: React.FC = () => {
  const { login } = useAuth();
  const [step, setStep] = useState<'phone' | 'otp'>('phone');
  const [phone, setPhone] = useState('');
  const [otp, setOtp] = useState('');
  const [isLoading, setIsLoading] = useState(false);

  const handleGetOtp = async () => {
    if (phone.length < 10) {
        toast.error("Enter valid 10-digit number");
        return;
    }
    setIsLoading(true);
    // Fast API simulation
    setTimeout(() => {
        setIsLoading(false);
        setStep('otp');
        toast.success(`OTP sent to ${phone}`);
    }, 500);
  };

  const handleVerify = async () => {
    if (otp.length !== 4) {
        toast.error("Enter 4-digit OTP");
        return;
    }
    setIsLoading(true);
    try {
        const success = await login(phone);
        if (!success) {
            // If login returns false, it means user not found (handled in context toast)
            setIsLoading(false);
        }
    } catch (e) {
        toast.error("An error occurred");
        setIsLoading(false);
    }
  };

```

A.1.2 Dashboard.tsx – Group Overview Dashboard

The dashboard screen displays a high-level summary of the SHG's financial health, including total members, total savings, active loans, and pending repayments. It consumes data from the shared data context and shows alerts and reminders for upcoming repayments and meetings. This screen acts as the primary landing page after login and serves as a navigation hub to other modules.

Code :

```tsx
export const Dashboard: React.FC<DashboardProps> = ({ onChangeView }) => {
  const { stats, transactions, refreshData, lastSynced, isSyncing, syncStatus, conflicts, resolveConflict,
  const { t, language } = useLanguage();
  const { currentUserRole, user } = useAuth();
  const [showConflictModal, setShowConflictModal] = useState(false);
  const [dismissedInsights, setDismissedInsights] = useState<string[]>([]);

  // Trigger AI analysis when language changes or data loads
  useEffect(() => {
     const timer = setTimeout(() => {
        generateAIInsights(language);
     }, 1000);
     return () => clearTimeout(timer);
  }, [language]);

  // Format last synced time
  const timeSinceSync = useMemo(() => {
    const diff = Math.floor((new Date().getTime() - lastSynced.getTime()) / 60000);
    if (diff < 1) return t('justNow');
    if (diff < 60) return `${diff}m ago`;
    return `${Math.floor(diff / 60)}h ago`;
  }, [lastSynced, t]);

  const handleRefresh = async () => {
    await refreshData();
    await generateAIInsights(language);
    toast.success(t('synced'), {
      icon: '☁️',
      style: { borderRadius: '10px', background: '#333', color: '#fff' },
    });
  };
```

A.2 Context and State Management

A.2.1 AuthContext.tsx – Authentication and Role State

This context manages global authentication state, including the current user, access token, and role (member, leader, animator, federation officer). It exposes functions for login, logout, and session refresh, and wraps the app so that any screen can read auth status and enforce role-based access to features.

Code :

```tsx
export const AuthProvider: React.FC<{ children: React.ReactNode }> = ({ children }) => {
  const [user, setUser] = useState<User | null>(null);
  const [isLocked, setIsLocked] = useState(true);
  const [settings, setSettings] = useState<AuthSettings>({
    isPinEnabled: false,
    pin: '',
    isBiometricEnabled: false,
    onboardingCompleted: false
  });

  const loadState = () => {
    try {
      const savedSettings = localStorage.getItem(AUTH_STORAGE_KEY);
      if (savedSettings) {
        setSettings(JSON.parse(decrypt(savedSettings)));
      }

      const savedUser = localStorage.getItem(USER_STORAGE_KEY);
      if (savedUser) {
        const parsedUser = JSON.parse(decrypt(savedUser));
        setUser(parsedUser);
      } else {
        setUser(null);
        setIsLocked(false);
      }
    } catch (e) {
      console.error("Auth load error", e);
      setIsLocked(false);
    }
  };
```

A.2.2 DataContext.tsx – Offline Data and Synchronization

The data context encapsulates the application's core business data: members, groups, transactions, meetings, and reports. It provides methods to fetch, add, and update records, and internally coordinates with local storage and sync routines. By centralising data access, it simplifies state management and ensures that UI components remain focused on presentation logic.

Code:

```
export const DataProvider: React.FC<{ children: React.ReactNode }> = ({ children }) => {
  // Initialize state from localStorage
  const [members, setMembers] = useState<Member[]>(() => {
    const saved = localStorage.getItem(STORAGE_KEY);
    if (saved) {
      try {
        const parsed = JSON.parse(saved);
        return parsed.members || [];
      } catch (e) { return []; }
    }
    return [];
  });

  const [transactions, setTransactions] = useState<Transaction[]>(() => {
    const saved = localStorage.getItem(STORAGE_KEY);
    if (saved) {
      try {
        const parsed = JSON.parse(saved);
        return parsed.transactions || [];
      } catch (e) { return []; }
    }
    return [];
  });

  const [meetings, setMeetings] = useState<Meeting[]>(() => {
    const saved = localStorage.getItem(STORAGE_KEY);
    if (saved) {
      try { const parsed = JSON.parse(saved); return parsed.meetings || []; } catch (e) { return []; }
    }
    return [];
  });

  const [loans, setLoans] = useState<Loan[]>(() => {
    const saved = localStorage.getItem(STORAGE_KEY);
    if (saved) {
        try { const parsed = JSON.parse(saved); return parsed.loans || []; } catch (e) { return []; }
    }
    return [];
  });
```

APPENDIX II

RESEARCH PROFESSIONAL PRACTICE

PATENT PUBLICATION :

TITLE : SOLAR POWERED AUTOMATED MULTITASKING AGRI SYSTEM

STATUS : PUBLISHED

सत्यमेव जयते

Office of the Controller General of Patents, Designs & Trade Marks
Department for Promotion of Industry and Internal Trade
Ministry of Commerce & Industry,
Government of India

INTELLECTUAL PROPERTY INDIA
PATENTS | DESIGNS | TRADE MARKS
GEOGRAPHICAL INDICATIONS

Application Details	
APPLICATION NUMBER	202441076824
APPLICATION TYPE	ORDINARY APPLICATION
DATE OF FILING	10/10/2024
APPLICANT NAME	**1 . SRI SAIRAM ENGINEERING COLLEGE** **2 . VAISHNAVI S** **3 . NIVEDITHA S** **4 . PREETHI S** **5 . M.NITHYA**
TITLE OF INVENTION	SOLAR POWERED AUTOMATED MULTITASKING AGRI SYSTEM
FIELD OF INVENTION	MECHANICAL ENGINEERING
E-MAIL (As Per Record)	
ADDITIONAL-EMAIL (As Per Record)	
E-MAIL (UPDATED Online)	
PRIORITY DATE	
REQUEST FOR EXAMINATION DATE	10/10/2024
PUBLICATION DATE (U/S 11A)	18/10/2024

PAPER PUBLICATION:

TITLE : Non network intensive record keeping system for SHG

STATUS : Presented the paper in the ICSDS Conference

PAPER PUBLICATION:

TITLE : Autonomous Solar Powered Farming and Monitoring unit

STATUS : Presented the paper in the ICSDS Conference

INTERNATIONAL CONFERENCE ON SUSTAINABLE DEVELOPMENTS IN COMPUTER ENGINEERING, GREEN TECHNOLOGY & SMART SYSTEMS (ICSDS-2025)

Certificate of Presentation

This is to certify that VAISHNAVI S of M.TECH CSE , SRI SAIRAM ENGINEERING COLLEGE ,TAMIL NADU, INDIA has presented his/her research paper titled "AUTONOMOUS SOLAR-POWERED FARMING AND MONITORING UNIT" in the ICSDS-2025 organized by Hooghly Engineering & Technology College, Hooghly, West Bengal held from December 20th to 21st, 2025.

PROF. (DR.) B. P. PATTANAIK
Conference Secretry

DR. D. K. SHUKLA
Conference Secretry

DR. SUMANTA DAW
Convener

PROJECT OUTCOMES [PO]

PO1:

To develop and validate an offline-first SHG management platform that ensures reliable capture and storage of financial and meeting records in low-connectivity environments.

PO2:

To implement and evaluate conflict-aware synchronization and role-based access control for maintaining consistent, secure SHG data across devices and hierarchy levels.

PO3:

To assess the impact of SHG Connect on record accuracy, reporting efficiency, and transparency for SHG members, leaders, and federation stakeholders.

Mapping Between Project Outcomes – Program Outcomes(PO) & Program Specific Outcomes (PSO)

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO1**	**PSO2**
Project Outcome 1	3	2	3	2	3	2	1	1	1	1	1	3	3
Project Outcome 2	3	3	3	2	3	2	2	2	2	2	2	3	3
Project Outcome 3	2	3	3	2	2	3	2	2	3	2	2	2	3

JUSTIFICATION

Project Outcome 1

"To develop and validate an offline-first SHG management platform that ensures reliable capture and storage of financial and meeting records in low-connectivity environments."

- **PO1 (3) – Engineering Knowledge:** Uses computing, databases, and mobile app fundamentals to design the offline-first architecture.
- **PO2 (2) – Problem Analysis:** Analyses existing SHG record-keeping issues and connectivity constraints to frame requirements.
- **PO3 (3) – Design/Development of Solutions:** Directly focused on designing and implementing a robust, offline-first solution.
- **PO4 (2) – Investigation:** Requires basic testing and validation of reliability and data integrity under different connectivity scenarios.
- **PO5 (3) – Engineering Tool Usage:** Uses modern tools (React/TS, local DB, APIs) and modelling to implement offline storage and sync.
- **PO6 (2) – Engineer & World:** Considers rural context, financial inclusion, and sustainability in designing the platform.
- **PO7 (1) – Ethics:** Touches privacy and secure handling of member financial data.
- **PO8 (1) – Teamwork:** Implemented through team-based development and coordination.
- **PO9 (1) – Communication:** Requires basic documentation and demonstration of the platform.
- **PO10 (1) – Project Management:** Involves planning milestones, tasks and basic resource management.
- **PO11 (1) – Life-long Learning:** Exposure to new offline-first patterns and libraries encourages continuous learning.
- **PSO1 (3):** Strong use of software design, data modelling and implementation for a full stack computational solution.
- **PSO2 (3):** Applies programming principles to a real social problem (rural SHGs).

Project Outcome 2

“To implement and evaluate conflict-aware synchronization and role-based access control for maintaining consistent, secure SHG data across devices and hierarchy levels.”

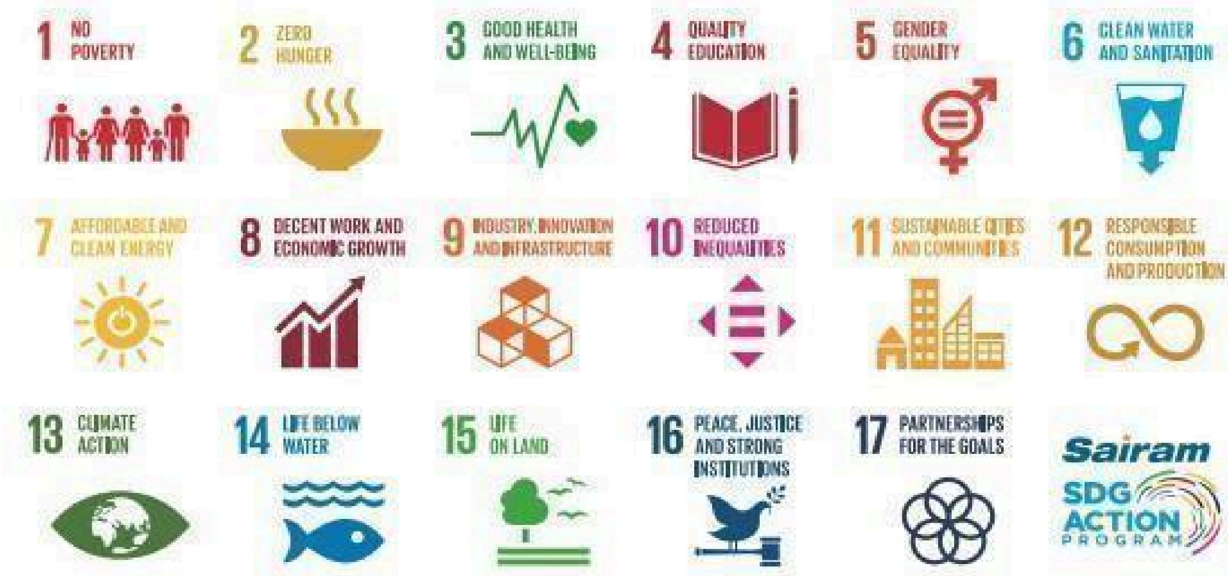
- **PO1 (3):** Applies core CS/IT knowledge (concurrency, consistency, security) to complex data-sync problems.
- **PO2 (3):** Requires detailed analysis of conflict scenarios, roles, and permissions before design.
- **PO3 (3):** Designs and develops concrete mechanisms like conflict resolution rules and RBAC.
- **PO4 (2):** Involves designing experiments and tests to verify sync correctness and security behaviour.
- **PO5 (3):** Heavy use of APIs, local/remote DBs, and dev tools to implement sync, queues, and access control.
- **PO6 (2):** Considers security, privacy, and fairness for different SHG roles and stakeholders.
- **PO7 (2):** Directly linked to ethics, confidentiality, and correct access to members’ financial records.
- **PO8 (2):** Developed collaboratively, especially for integration between frontend, backend, and context layers.
- **PO9 (2):** Requires clear communication of sync rules, error messages, and documentation for users and evaluators.
- **PO10 (2):** Managing this technical core of the project needs planning, risk handling, and division of responsibilities.
- **PO11 (2):** Demands learning emerging patterns in offline sync, security libraries, and critical thinking around failure cases.
- **PSO1 (3):** High-level application of design and testing skills to ensure correct computational behaviour.
- **PSO2 (3):** Strong analytical reasoning to handle conflicts, security.

Project Outcome 3

"To assess the impact of SHG Connect on record accuracy, reporting efficiency, and transparency for SHG members, leaders, and federation stakeholders."

- **PO1 (2):** Uses analytical knowledge and basic statistics to interpret accuracy and efficiency improvements.
- **PO2 (3):** Involves analysing data before/after using the app and drawing substantiated conclusions.
- **PO3 (3):** Evaluates whether the designed solution meets user needs for better reporting and transparency.
- **PO4 (2):** Requires small-scale studies, user feedback, or experiments to collect and interpret evaluation data.
- **PO5 (2):** Uses tools such as spreadsheets, dashboards, or analytics libraries to measure performance metrics.
- **PO6 (3):** Directly addresses societal impact—transparency, inclusion, and governance in rural SHGs.
- **PO7 (2):** Considers ethical handling of user data during evaluation and honest reporting of results.
- **PO8 (2):** Evaluation is carried out by the project team in coordination with SHG-like users or testers.
- **PO9 (3):** Strong emphasis on presenting findings via reports, charts, and presentations for different stakeholders.
- **PO10 (2):** Includes planning and managing evaluation activities as mini-projects.
- **PO11 (2):** Encourages reflective learning about limitations and future improvements based on findings.
- **PSO1 (2):** Uses computational outputs (reports, logs) to analyse system performance and correctness.
- **PSO2 (3):** Strong link to assessing social and environmental impact

Imagine the Future and
Make it happen!



Together let's build a better world where there is NO POVERTY and ZERO HUNGER.
We have GOOD HEALTH AND WELL BEING QUALITY EDUCATION and full GENDER EQUALITY everywhere.
There is CLEAN WATER AND SANITATION for everyone. AFFORDABLE AND CLEAN ENERGY
which will help to create DECENT WORK AND ECONOMIC GROWTH. Our prosperity shall be fuelled
by investments in INDUSTRY, INNOVATION AND INFRASTRUCTURE that will help us to
REDUCE INEQUALITIES by all means. We will live in SUSTAINABLE CITIES AND COMMUNITIES.
RESPONSIBLE CONSUMPTION AND PRODUCTION will help in healing our planet.
CLIMATE ACTION will reduce global warming and we will have abundant,
flourishing LIFE BELOW WATER, rich and diverse LIFE ON LAND.
We will enjoy PEACE AND JUSTICE through STRONG INSTITUTIONS
and will build long term PARTNERSHIPS FOR THE GOALS.

For the goals to be reached, everyone needs to do their part: governments, the private sector, civil society and **People like you.**

Together we can...

Sai Prakash LeoMuthu

Chairman & CEO – Sairam Institutions